

Fusion Constructive Realism: A Compute Paradigm

FCR Research Collaboration

August 9, 2026

Abstract

We formulate the computational infrastructure for *Fusion Constructive Realism* (FCR) using algebraic representation theory, group word metrics, and spectral transfer operators. By replacing naive numeric $N \times N$ matrix stepping with discrete symmetry cosets G/H , metric distances $\mu(g_1, g_2) = d_G(g_1^{-1}g_2)$ are evaluated algebraically in $O(1)$ memory. State deduplication on the Quotient DAG reduces from $O(N^3)$ graph canonicalization to $O(1)$ group character comparisons. Evolution on the Quotient DAG is formulated as a sparse transfer operator $\mathbf{T}$, allowing Born-rule display probability distributions $P(d_k)$ to be computed directly via spectral decomposition. Finally, we show that Class-2 particle limit cycles $(\mathcal{M}_3, \mathcal{M}_6, \mathcal{M}_{12})$ correspond to irreducible representations (irreps) of polyhedral automorphism groups, enabling analytic mass hierarchy calculations without step-by-step numerical simulation loops.

1 Algebraic State Space & Group Word Metrics

Representing primitive metric distances $\mu(p, q)$ as unconstrained real-valued matrices scales as $O(N^2)$ in memory, rendering large-scale graph simulations intractable. In FCR, background lattices $\mathcal{M}_0$ and particle frames $\mathcal{M}_n$ exhibit local discrete symmetries.

Definition 1 (Coset State Representation). *Points $p \in P$ are indexed as elements of a discrete symmetry group or coset space G/H . The primitive metric distance $\mu(p, q)$ is defined algebraically via the group word metric:*

$$\mu(p, q) \equiv d_G(p^{-1}q) = \min\{k \mid p^{-1}q = s_1 s_2 \dots s_k, s_i \in S\} \quad (1)$$

where S is the minimal generator set of G .

This algebraic formulation replaces $O(N^2)$ matrix storage with $O(1)$ generator lookups.

Theorem 1 ($O(1)$ Character-Based Graph Deduplication). *Two universe states Q_a, Q_b defined on group symmetry G are isomorphic ($Q_a \cong Q_b$) if and only if their correlation character profiles match:*

$$\chi_{Q_a}(g) = \chi_{Q_b}(g) \quad \forall g \in C(G) \quad (2)$$

where $C(G)$ is the set of conjugacy classes of G . Canonical hashing on the Quotient DAG is evaluated in $O(|C(G)|) \sim O(1)$ time rather than via $O(N^3)$ graph colorings.

2 Transfer Operator Algebra on the Quotient DAG

Temporal evolution across poset steps $T \rightarrow T + \tau$ is governed by the global **Spectral Transfer Operator** $\mathbf{T}$, whose matrix elements T_{ab} carry the complex correlation phase transitions driven by Triangulation (Mutation 2) and Decay (Mutation 3).

Definition 2 (Transfer Operator $\mathbf{T}$). For states $Q_a, Q_b \in \mathcal{M}$, the transfer matrix element T_{ab} is:

$$T_{ab} = \begin{cases} \prod_{e \in \text{mut}(Q_a \rightarrow Q_b)} C(e) \cdot e^{-\gamma_d \mu(e)}, & \text{if } Q_a \rightarrow Q_b \text{ is a valid 1-step transition} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Spectral Evaluation of the Born Rule: Path counting over display equivalence classes $\mathcal{O}_k$ avoids explicit path enumeration. The total phase-weighted trajectory amplitude after τ steps is given by matrix powers:

$$Z(Q_0 \rightarrow \mathcal{O}_k; \tau) = \sum_{Q \in \mathcal{O}_k} (\mathbf{T}^\tau)_{0,Q} \quad (4)$$

Display probabilities $P(d_k)$ are extracted by projecting onto the dominant eigenvectors $|\lambda_j\rangle$ of $\mathbf{T}$ via Lanczos iteration:

$$P(d_k) = \frac{|\langle d_k | \mathbf{T}^\tau | Q_0 \rangle|^2}{\sum_j |\langle d_j | \mathbf{T}^\tau | Q_0 \rangle|^2} \quad (5)$$

3 Soliton Classification via Irreducible Representations

Rather than searching for stable particles via numerical trial-and-error, Class-2 Topodynamical Attractors are identified as ****irreducible representations (irreps)**** of discrete polyhedral groups under the capacity constraint $\mathcal{C}_{\text{ext}} \leq 1$.

Definition 3 (Irrep Soliton Criterion). A subgraph $\mathcal{M}_n$ is a stable Class-2 limit cycle if its correlation phase character $\chi_{\mathcal{M}_n}$ satisfies orthogonality and capacity unitarity:

$$\frac{1}{|G|} \sum_{g \in G} |\chi_{\mathcal{M}_n}(g)|^2 \leq \mathcal{C}_{\text{ext}} = 1 \quad (6)$$

- **Electron** ($\mathcal{M}_3$): Irreducible character of the cyclic permutation group $\mathbb{Z}_3$.
- **Muon** ($\mathcal{M}_6$): Irreducible character of the octahedral point group O_h ($|O_h| = 48$).
- **Tau** ($\mathcal{M}_{12}$): Irreducible character of the icosahedral point group I_h ($|I_h| = 120$).

Lepton rest masses are calculated analytically as the ratio of character norms weighted by the algebraic Laplacian spectrum $\lambda(\mathbf{L}_{G/H})$, bypassing time-stepping simulations entirely.

4 Algebraic Capacity Audit & Sub-Tick Virtual Lock

Capacity overload ($\sum_{q \neq p} |C(p, q)|^2 > 1$) triggers sub-tick Virtual Lock ($\mathcal{M}_{1/2}, \Delta T = 0$). In the algebraic paradigm, this corresponds to ****character decomposition into non-unitary channels****:

[Algebraic Virtual Lock Resolution] 1. For group state Q_T , compute local character norm $N(p) = \sum_{g \in G} |\chi_p(g)|^2$. 2. If $N(p) > 1$, decompose χ_p into irreducible components:

$$\chi_p = a_1 \rho_1 \oplus a_2 \rho_2 \oplus \cdots \oplus a_k \rho_k \quad (7)$$

3. Eject non-unitary irrep components carrying excess norm $N(p) - 1.0$ as virtual half-edges $\mathcal{M}_{1/2}$. 4. Project remaining character onto substrate representation $\mathcal{M}_0$. Re-evaluate until $N(p) \leq 1.0 \quad \forall p$.

Table 1: Computational Abstractions in the Algebraic FCR Engine

Physical Concept	Numeric Engine (Naive)	Algebraic Engine (Proposed)
Metric Space (P, μ)	$N \times N$ Matrix μ_{ij}	Group Word Metric $d_G(g_1^{-1}g_2)$
State Hashing	Nauty $O(N^3)$ Coloring	Group Character Matching $O(1)$
Time Evolution	Poset Tick Loop $Q_T \rightarrow Q_{T+1}$	Sparse Transfer Operator $\mathbf{T}^\tau$
Born Rule	Path Traversal on DAG	Eigenvector Projection $\langle d_k \mathbf{T}^\tau Q_0 \rangle$
Particle Discovery	Random Seed Injection	Irrep Classification of O_h, I_h

5 Software Architecture & Sparse Spectral Solvers

The algebraic compute engine relies on sparse eigensolver libraries (ARPACK / Spectra) and group character tables rather than explicit graph adjacency lists:

This algebraic formulation guarantees that $P(d_k)$ and particle mass spectra are computable with $O(\text{poly}(\log N))$ memory efficiency.